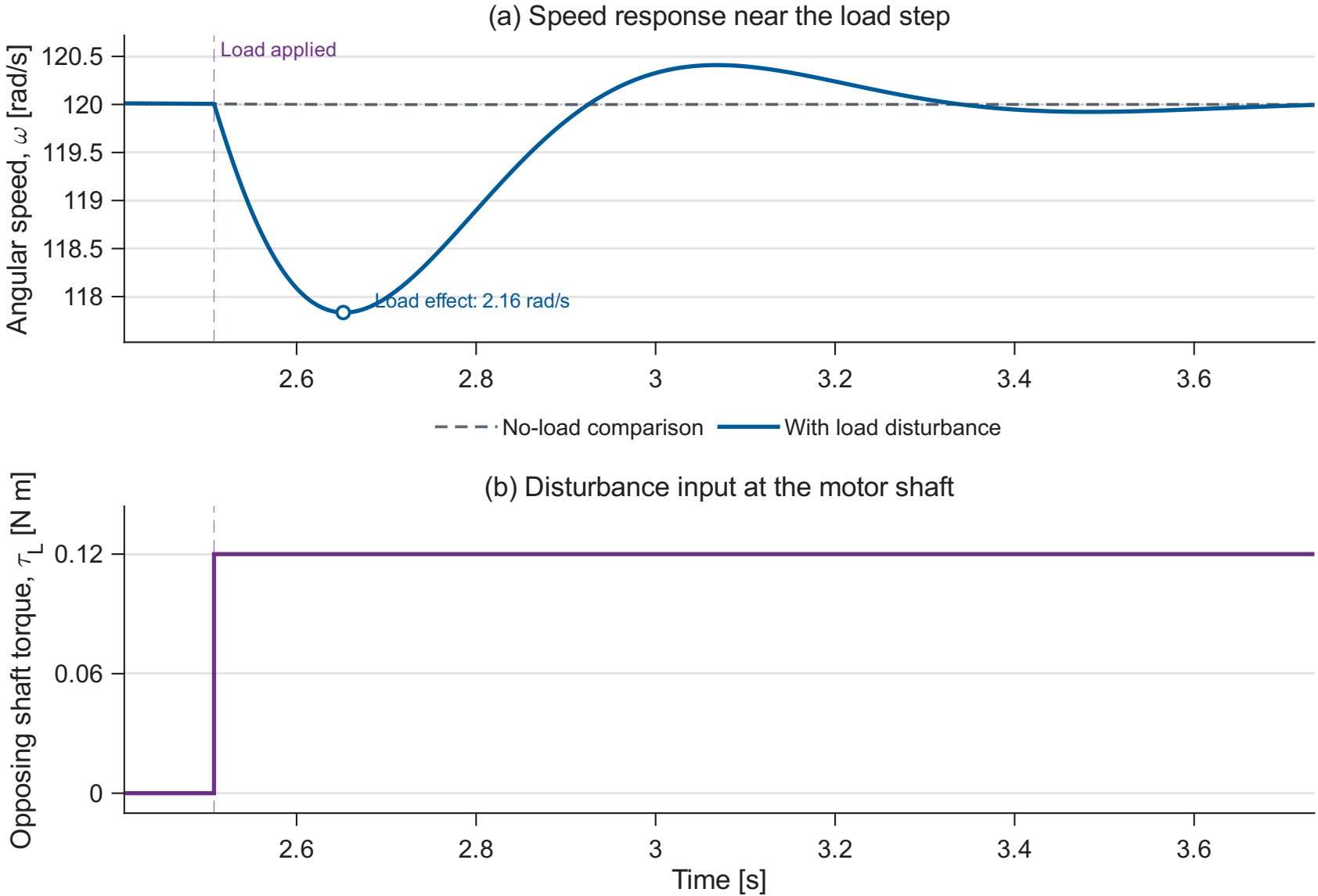
(a) Speed response near the load step
Load applied
Load effect: 2.16 rad/s
Angular speed, ω [rad/s]
120.5
120
119.5
119
118.5
118
2.6
2.8
3
3.2
3.4
3.6
No-load comparison
With load disturbance
(b) Disturbance input at the motor shaft
Opposing shaft torque, τL [N m]
0.12
0.06
0
2.6
2.8
3
3.2
3.4
3.6
Time [s]